

FRANKFURT AM MAIN, Germany

WMO: 106370

Lat: 50.0258N

Lon: 8.5214E

Elev: 104

StdP: 100.08

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.2	(c) -5.9	(d) -13.2	(e) 1.2	(f) -6.4	(g) -10.2	(h) 1.6	(i) -4.1	(j) 12.4	(k) 9.6	(l) 11.0	(m) 8.7	(n) 3.1	(o) 40	(p) 0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	32.1	20.0	30.0	19.2	28.0	18.4	21.2	28.6	20.4	27.0	19.6	25.7	3.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.1	22.4	18.2	13.3	21.9	17.4	12.6	21.7	61.8	28.6	59.1	26.8	56.4	25.8	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.6	8.4	7.4	-11.1	35.5	3.4	1.5	-13.6	36.6	-15.5	37.4	-17.4	38.3	-19.9	39.3		
			-11.7	22.8	3.3	0.7	-14.1	23.3	-16.0	23.7	-17.9	24.1	-20.2	24.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	2.0	3.0	6.7	11.2	14.7	18.6	20.8	19.8	15.8	11.2	6.3	3.2
	DBStd	7.56	4.47	4.09	3.95	4.01	3.78	3.63	3.50	3.28	3.14	3.61	3.69	4.19
	HDD10.0	961	249	196	114	33	4	0	0	0	1	30	120	213
	HDD18.3	2854	507	428	360	214	124	42	15	21	89	223	362	470
	CDD10.0	1379	1	1	13	70	151	258	334	304	174	66	8	2
	CDD18.3	233	0	0	0	1	12	49	91	67	13	1	0	0
	CDH23.3	2157	0	0	0	32	132	452	849	584	103	4	0	0
	CDH26.7	728	0	0	0	2	28	143	329	202	23	0	0	0
Wind	WSAvg	3.4	3.8	3.8	3.9	3.5	3.5	3.4	3.3	3.0	3.1	3.2	3.3	3.6
Precipitation	PrecAvg	692	54	52	45	42	62	64	72	61	56	59	59	64
	PrecMax	851	137	130	122	89	123	134	163	121	134	171	99	177
	PrecMin	497	4	8	8	0	16	18	30	3	14	7	1	12
	PrecStd	96	26	30	27	21	29	28	32	32	28	33	25	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	14.2	20.2	26.2	29.5	33.0	35.1	34.8	29.8	23.8	16.8	13.1
		MCWB	10.7	9.3	11.8	14.7	19.0	20.5	20.9	20.7	19.0	16.8	13.4	10.5
	2%	DB	11.6	12.0	17.5	24.0	26.8	30.1	32.9	31.1	26.2	20.1	14.2	11.9
		MCWB	9.4	8.6	10.6	13.9	17.5	19.4	20.1	19.2	18.0	15.6	11.8	9.9
	5%	DB	9.8	10.4	15.1	21.1	24.2	28.0	30.2	28.9	23.8	18.1	13.0	10.4
		MCWB	8.2	7.8	9.8	13.1	15.8	18.6	19.1	18.7	17.2	14.7	10.9	8.7
	10%	DB	8.1	9.0	13.0	18.9	22.1	25.9	28.0	26.8	21.6	16.8	11.6	9.0
		MCWB	6.8	6.8	9.1	12.1	14.6	17.9	18.4	18.2	16.1	14.0	10.0	7.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	10.7	13.0	16.0	20.3	22.4	22.1	21.9	20.4	17.6	14.2	11.5
		MCDB	12.8	12.9	17.3	22.7	27.9	30.0	31.5	29.0	25.8	21.1	15.7	12.7
	2%	WB	9.7	9.2	11.7	14.6	18.5	20.7	21.1	20.9	19.1	16.3	12.6	10.2
		MCDB	11.0	11.2	15.7	21.8	24.9	27.5	29.0	28.0	24.5	19.2	14.1	11.5
	5%	WB	8.1	8.1	10.6	13.5	16.9	19.6	20.4	19.9	17.9	15.3	11.2	9.0
		MCDB	9.4	10.1	14.0	19.7	22.5	25.9	27.1	26.1	22.4	17.7	12.7	10.4
	10%	WB	6.7	6.9	9.4	12.6	15.6	18.7	19.6	19.1	16.9	14.3	10.1	7.6
		MCDB	7.9	8.7	12.7	17.9	20.2	24.5	25.6	24.5	20.8	16.3	11.3	8.7
Mean Daily Temperature Range	5% DB	MDBR	4.8	6.2	8.5	10.3	10.1	10.2	10.5	10.3	9.3	7.7	5.0	4.5
		MCDBR	6.0	8.8	12.2	14.2	13.8	13.8	14.5	14.2	12.6	9.9	6.7	5.6
		MCWBR	4.7	6.0	6.7	6.4	6.0	5.4	4.4	5.1	5.7	5.7	4.8	4.5
	5% WB	MCDBR	5.7	7.7	10.4	12.3	12.2	12.1	12.2	11.9	10.4	8.7	6.0	5.5
		MCWBR	4.8	5.6	6.0	5.9	5.8	5.3	4.4	5.1	5.1	5.4	4.7	4.7
Clear-Sky Solar Irradiance	taub	0.331	0.360	0.397	0.415	0.420	0.422	0.427	0.421	0.392	0.384	0.356	0.329	
	taud	2.363	2.296	2.208	2.211	2.231	2.252	2.234	2.277	2.339	2.330	2.346	2.359	
	Ebn at Noon	698	764	796	826	839	838	827	812	797	725	654	639	
	Edh at Noon	67	91	117	130	133	131	132	120	102	87	68	59	
All-Sky Solar Radiation	RadAvg	0.79	1.53	2.67	4.21	4.97	5.53	5.23	4.51	3.28	1.92	0.90	0.60	
	RadStd	0.12	0.24	0.38	0.54	0.54	0.49	0.58	0.42	0.43	0.20	0.11	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page